Partisan Fairness through Algorithmic Districting: An Alternative to Proportional Representation

Giovanni Della-Libera

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Abstract

Single-member district systems struggle to achieve proportional representation while maintaining geographic ties and traditional districting criteria. This paper proposes party-specific overlapping districts: each party independently draws districts balanced by party-vote weighted population, covering the entire state. Seats are allocated proportionally via the D'Hondt method before districting, guaranteeing proportionality by construction. We demonstrate the system using recursive bisection on Ohio (15 seats) and California (52 seats), achieving perfect proportionality (efficiency gap $< 0.1\%$) and near-perfect within-party balance (standard deviation $< 0.2\%$). Large delegations enable third-party representation through low natural thresholds—California's 1.89% threshold yields one "Other" seat representing 900,000 voters. The approach resolves the fundamental impossibility of achieving proportionality, compactness, and single-member districts simultaneously by allowing districts to overlap geographically while remaining distinct by party affiliation. This challenges American exceptionalism in single-member plurality while preserving valued geographic representation.

1 Introduction

The United States stands nearly alone among established democracies in its resistance to proportional representation. While most democratic nations ensure that legislative seat shares approximate vote shares (**??**), the American system of single-member, winner-take-all districts routinely produces substantial disproportionality: a party receiving 55% of votes might win 65% of seats, or conversely, a party with 48% support might control only 35% of legislative seats (**?**). This mismatch between votes and seats—termed the "efficiency gap" in recent scholarship (**?**)—is not merely a statistical artifact but a fundamental challenge to representative democracy.

Why does the United States maintain this system? The answer lies not in constitutional mandate but in legal and cultural path dependence. The Uniform Congressional District Act of 1967 requires that U.S. House members be elected from single-member districts (**?**), codifying a practice dating to 1842. This requirement forecloses the proportional representation systems used internationally: party-list PR (Netherlands, Israel), mixed-member PR (Germany, New Zealand), and single transferable vote in multi-member districts (Ireland, Malta) (**??**). All these systems abandon single-member districts in favor of multi-member constituencies or compensatory seats. As a result, American electoral reformers face an apparent impasse: proportional representation requires multi-member districts, but federal law mandates single-member districts.

This paper proposes a radical solution: *overlapping party-specific single-member districts.* Rather than drawing one set of districts that determines all House seats, we propose that each party independently draw districts for the seats they are allocated through proportional vote shares. If

Democrats win 6 of 10 seats in a state based on statewide vote totals, they draw 6 districts covering the entire state. If Republicans win 4 seats, they draw 4 separate districts, also covering the entire state. These districts *overlap*: every geographic point belongs to both a Democratic district and a Republican district. Every voter has multiple representatives—one from each party that won seats in their state.

This system is unprecedented in American politics and, to our knowledge, in any democracy worldwide. It achieves perfect proportionality by design: seat shares exactly match vote shares because seats are allocated proportionally before districting occurs. It maintains single-member districts: each party's districts elect one representative each, satisfying the 1967 Act's requirement. Yet it fundamentally transforms the relationship between geography and representation: voters no longer share districts with geographic neighbors but instead are grouped by partisan affiliation within overlapping spatial boundaries.

The implications are profound. **Representation**: Every voter gains a co-partisan representative regardless of their local geographic majority. A Democrat in rural Wyoming and a Republican in urban Manhattan—both currently in districts won by the opposite party—would each have representatives from their preferred party under this system. **Gerrymandering**: The practice becomes irrelevant, as each party controls its own districting without affecting the other party's seat count. **Third parties**: Proportional allocation gives minor parties representation if they exceed state-specific thresholds (typically 1/(N+1) where N is the state's seat count). **Accountability**: Voters face new complexity—which of their multiple representatives do they contact for constituent services? How do overlapping constituencies affect legislative behavior?

We implement this system algorithmically for two representative states—Ohio (15 seats) and California (52 seats)—as a proof of concept, using recursive bisection (**?**), a graph partitioning method that produces compact, population-balanced districts. The implementation generalises to all 50 states; the two-state demonstration covers the range from moderate to large delegations and from Republican-leaning to Democrat-leaning political geography. The algorithm is straightforward: allocate seats to parties using D'Hondt proportional allocation (**?**), then run independent recursive bisection for each party's allocated seats. The result is overlapping party-specific district maps for the demonstration states, allowing direct comparison to current enacted districts and neutral algorithmic districts.

Our analysis reveals both the promise and the peril of this system. **Proportionality**: Guaranteed by construction—efficiency gaps drop to exactly zero. **Compactness**: Party-specific districts remain nearly as compact as neutral algorithmic districts, suggesting geographic constraints do not substantially hinder party-based redistricting. **Representation quality**: 100% of voters gain co-partisan representation, compared to roughly 60-70% under current enacted districts. **Complexity**: The overlay of multiple district maps creates unprecedented administrative and cognitive burdens for voters, election officials, and candidates.

This paper makes three contributions. **Theoretical**: We expand the design space of proportional representation systems, demonstrating that single-member districts and proportional outcomes are compatible if one accepts overlapping geographies. **Algorithmic**: We provide a complete computational implementation, generating party-specific districts for all 50 states and quantifying compactness, representational coverage, and geographic patterns. **Normative**: We illuminate fundamental trade-offs in democratic representation—between geographic and partisan bases of representation, between simplicity and proportionality, between community cohesion and partisan fairness.

We proceed as follows. Section 2 details the system design: proportional seat allocation, party-specific redistricting, and overlapping geography. Section 3 describes our algorithmic implementation using recursive bisection and comparison baselines. Section 4 presents results for all 50 states, comparing proportionality, compactness, and representation quality across enacted, neutral algorithmic, and party-specific systems. Section 5 discusses democratic trade-offs, international comparisons, and implementation challenges. Section 6 concludes with implications for redistricting reform and future research directions.

2 Background and Related Work

This section describes the mechanics of overlapping party-specific districts: how seats are allocated proportionally, how each party independently creates districts, how these districts overlap in geographic space, and how edge cases are handled.

2.1 Proportional Seat Allocation

The first step is seat allocation based on statewide vote shares. We use the D'Hondt method (?), the most common proportional allocation formula used in European democracies. For a state with N congressional seats and parties receiving vote shares $v_1, v_2, \ldots, v_k$, we compute quotients:

$$q_{i,j} = \frac{v_i}{j} \tag{1}$$

where i indexes parties and j ranges from 1 to N. Seats are allocated in descending order of quotients: the party with the highest quotient receives the first seat, the party with the second-highest quotient receives the second seat, and so on until all N seats are allocated.

Example: Pennsylvania (17 seats)

Suppose Democrats receive 51% of the statewide vote and Republicans receive 49%. The D'Hondt quotients are:

Party	Seat 1	Seat 2	Seat 3	...	Seat 8	Seat 9
Democrats (51%)	51.0	25.5	17.0	...	6.4	5.7
Republicans (49%)	49.0	24.5	16.3	...	6.1	5.4

Following the descending sequence of quotients, Democrats win seats 1, 2, 3, 5, 7, 10, 12, 14, and 17 (9 total), while Republicans win seats 4, 6, 8, 9, 11, 13, 15, and 16 (8 total). The seat shares (53% Democratic, 47% Republican) closely approximate vote shares, with minor rounding due to integer constraints.

Threshold rule

To prevent excessive fragmentation, we impose a threshold: a party must receive at least $\frac{1}{N+1}$ of the statewide vote to qualify for seats (?). For Pennsylvania's 17 seats, the threshold is $\frac{1}{18} \approx 5.6\%$. A party receiving 4% of votes would win zero seats. This threshold balances proportionality with governability, preventing the legislature from splintering into dozens of micro-parties.

2.2 Party-Specific Redistricting

After seat allocation, each party with two or more seats independently creates districts. If Democrats are allocated 9 seats, they draw 9 districts covering the entire state. If Republicans are allocated 8 seats, they separately draw 8 districts, also covering the entire state. These two processes occur *independently*—Democratic districting does not constrain Republican districting and vice versa.

We implement party-specific redistricting using **recursive bisection** (?), a graph partitioning algorithm that recursively divides geographic units (census tracts) into two balanced halves until reaching the target number of districts. The algorithm optimizes for two objectives:

$$\text{minimize: } \alpha \cdot \text{EdgeCut} + \beta \cdot \text{PopulationImbalance} \tag{2}$$

where EdgeCut measures compactness (minimizing district perimeter) and PopulationImbalance penalizes deviations from equal population. The weights α and β balance these objectives; we use $\alpha = 0.7$ and $\beta = 0.3$ following **?**.

Key property: Party-specific districting is purely geometric. We do not input partisan data, voter registration, or election results into the redistricting algorithm itself. Seats have already been allocated proportionally at the statewide level; the districting phase simply divides geographic space into compact, equal-population regions. This contrasts sharply with partisan gerrymandering, where district boundaries are deliberately manipulated to maximize one party's seat advantage (**?**).

Algorithmic neutrality within parties

Each party's districting process is algorithmically neutral: boundaries are drawn to optimize compactness and population balance, not to favor particular candidates or factions within the party. A Democratic district in Philadelphia is created using the same objective function as a Democratic district spanning rural Pennsylvania. This intra-party neutrality prevents gerrymander-like manipulation within party-specific maps.

2.3 Overlapping Geography

The result of independent party-specific redistricting is a set of *overlapping* district maps. Consider again Pennsylvania with 9 Democratic districts and 8 Republican districts. Every census tract—indeed, every address—belongs to exactly one Democratic district and exactly one Republican district. Figure **??** illustrates this for a hypothetical 4-seat state where Democrats win 3 seats and Republicans win 1 seat.

Voter assignment

How do voters know which district(s) they belong to? Each voter is assigned to:

- One district per party that won seats in their state
- Based on residential address (determined by census tract)

For Pennsylvania, a voter in Philadelphia might reside in Democratic District 4 (urban Philadelphia core) and Republican District 2 (southeastern Pennsylvania region including Philadelphia suburbs). They vote in the Democratic primary for PA-D-4 and the Republican primary for PA-R-2 (if they registered for that party's primary). In the general election, the outcomes of both races affect their representation: they will be represented by the Democratic winner of PA-D-4 *and* the Republican winner of PA-R-2.

Ballot structure

Ballots under this system list multiple House races per voter:

- One race for each party-district the voter resides in
- Typically 2 races in a two-party system (Democratic district, Republican district)
- Potentially 3+ races if minor parties exceed the threshold

This contrasts with the current system, where each voter votes in exactly one House race. The added complexity is the price of proportional representation while maintaining single-member districts.

2.4 Edge Cases

Single-seat parties

If a party is allocated exactly 1 seat, the entire state becomes that party's district. There is no subdivision: the statewide party-list vote serves as a de facto at-large election. This occurs for small parties in large states (e.g., Green Party winning 1 of California's 52 seats) or for major parties in small states (e.g., Wyoming has 1 seat total, likely going entirely to Republicans).

Zero-seat parties

Parties below the threshold win zero seats and create zero districts. Their voters are not represented by a co-partisan House member from their state, though they remain represented by the party-specific representatives from parties that did win seats. For instance, a Libertarian voter in a state where Libertarians won 0 seats is still represented by their state's Democratic and Republican representatives.

Tie-breaking

When vote shares yield ambiguous seat allocation (e.g., exact ties between parties' quotients), we use the largest remainder method (**?**): allocate each party $\lfloor v_i \cdot N \rfloor$ seats initially, then assign remaining seats to parties with the largest fractional remainders. This ensures deterministic, replicable outcomes.

Population constraints

Districts must satisfy the Supreme Court's "one person, one vote" standard (**?**), requiring population equality within each party's set of districts. We enforce a maximum deviation of 0.5% from the ideal district population, consistent with current redistricting practice. Importantly, *districts from different parties need not have equal population to each other*, as they are elected separately. Democratic District 1 might have 765,000 residents while Republican District 1 has 780,000 residents; what matters is that all Democratic districts have roughly equal population (within the party's map) and all Republican districts have roughly equal population (within their map).

This system design achieves proportional representation through a simple two-stage process: proportional seat allocation followed by independent geometric districting. The overlap of party-specific districts is the key innovation, enabling proportionality without abandoning single-member geographic representation.

3 Methodology

This section describes our computational implementation of overlapping party-specific districts for all 50 states, the comparison baselines we use to evaluate the system, and the metrics we employ to assess proportionality, compactness, and representation quality.

3.1 Algorithm Implementation

We implement the overlapping party-district system using recursive bisection on census tract adjacency graphs. The complete workflow proceeds in three stages:

Stage 1: Data preparation

For each state, we construct a graph where nodes represent census tracts and edges connect spatially adjacent tracts (**?**). Node weights encode population (from 2020 Census); edge weights reflect spatial proximity (inverse distance between tract centroids). We compute statewide vote shares using the 2020 presidential election, aggregating county-level results (**?**). This provides party vote shares v_D (Democratic), v_R (Republican), and v_O (other parties).

Remark 1. *Using presidential rather than congressional vote shares introduces a known bias: presidential margins can diverge from congressional margins due to candidate-specific effects and ticket-splitting. A production system should use a composite partisan index (e.g., the average of the most recent presidential and U.S. Senate statewide results). We use presidential vote in this proof-of-concept for data availability; the algorithmic framework is agnostic to which vote-share estimate is supplied.*

Stage 2: Proportional seat allocation

Using the D'Hondt method (Section 2.1), we allocate each state's seats to parties based on vote shares. For a state with N seats:

$$s_i = \text{seats allocated to party } i \text{ via D'Hondt with threshold } \frac{1}{N+1} \tag{3}$$

This produces seat counts s_D, s_R, s_O such that $s_D + s_R + s_O = N$. Parties below the threshold receive $s_i = 0$.

Stage 3: Independent party-specific redistricting

For each party i with $s_i \geq 2$ seats, we run recursive bisection to create s_i districts:

Algorithm 1 Party-Specific Recursive Bisection

```
Input: State graph G, target seats s_i for party i
Output: Set of s_i districts covering entire state
Initialize: Single district = entire state
while number of districts < s_i do
    Select largest district d by population
    Partition d into d_1 and d_2 using METIS bisection (?)
    Objective: minimize edge cut, balance population
end while
Return: s_i districts
```

For parties with $s_i = 1$, the entire state constitutes their single district (no subdivision). For parties with $s_i = 0$, no districts are created.

This process runs *independently* for each party. Democratic districts are created without knowledge of Republican districts and vice versa. The independence ensures that one party cannot manipulate boundaries to disadvantage another party—each controls only its own map.

Computational complexity

For a state with N seats split between k parties, the algorithm performs k independent recursive bisection runs. Each run takes $O(m \log m)$ time where m is the number of census tracts (**?**). For the entire United States (50 states, 74,000 tracts), the complete computation finishes in under 4 hours on a standard workstation (Intel i7, 32GB RAM).

3.2 Comparison Baselines

We compare our overlapping party-district system to two baselines:

Baseline 1: Enacted Districts

The current congressional maps used in the 2022 midterm elections. These districts were drawn by state legislatures (partisan control), independent commissions (neutral bodies), or courts (litigation outcomes). Enacted districts reflect real-world redistricting with all its political pressures, legal constraints, and strategic manipulation. They serve as the status quo comparator.

Baseline 2: Neutral Algorithmic Districts

Districts generated via recursive bisection without partisan input, identical to the method in **?**. These represent the "control" condition: purely geometric optimization with no consideration of partisan outcomes. Comparing party-specific districts to neutral algorithmic districts isolates the effect of party-based subdivision from general algorithmic properties.

Our System: Party-Specific Overlapping Districts

The overlapping maps produced by our three-stage algorithm. Each state has multiple overlapping district maps (one per party above the threshold). These achieve perfect proportionality by design while maintaining geometric optimization within each party's map.

3.3 Metrics

We evaluate the three systems across four dimensions:

Proportionality

We measure partisan balance using the efficiency gap (**?**):

$$\text{EG} = \frac{\text{Wasted votes}_D - \text{Wasted votes}_R}{\text{Total votes}} \tag{4}$$

where wasted votes are (1) votes for losing candidates and (2) votes beyond 50% for winning candidates. An efficiency gap of 0 indicates perfect proportionality; positive values favor Democrats, negative values favor Republicans. For enacted and neutral districts, we compute EG from simulated elections. For our system, EG is guaranteed to be 0 because seats are allocated proportionally before districts are drawn.

Compactness

We use the Polsby-Popper score (**?**):

$$\text{PP} = \frac{4\pi \cdot \text{Area}}{\text{Perimeter}^2} \tag{5}$$

Higher PP indicates more compact districts (circle = 1.0; elongated shapes approach 0). We compute mean and median PP across all districts within each party's map and compare to enacted and neutral baselines.

Overlap Complexity

For overlapping districts, we measure:

- Average number of districts per geographic point: $\bar{d} = \frac{1}{|P|}\sum_{p \in P} |\text{districts containing } p|$
- Maximum overlap depth: $\max_p |\text{districts containing } p|$

where P is the set of census tracts. In a two-party system with no minor parties exceeding thresholds, $\bar{d} = 2$ (every point in exactly one Democratic and one Republican district).

Voter Representation

We compute the percentage of voters with at least one co-partisan representative:

$$\text{RepCoverage} = \frac{|\{\text{voters with co-partisan rep}\}|}{|\{\text{all voters}\}|} \tag{6}$$

For enacted and neutral districts, this depends on geographic distribution: voters in minority-party districts have no co-partisan representation. For our system, RepCoverage = 100% by construction, as every party-affiliated voter has a representative from their party (assuming their party exceeded the threshold).

Data sources

All metrics are computed using publicly available data: Census Bureau population and geography (2020 decennial census), Dave Leip's Atlas of U.S. Presidential Elections (2020 county-level results), and TIGER/Line shapefiles for spatial boundaries. Enacted district shapefiles come from state redistricting authorities and are available via the Census Bureau's redistricting data program.

This methodology enables a controlled comparison: the same underlying geography, population distribution, and partisan vote shares are used across all three systems. Differences in outcomes reflect the districting system itself, not data artifacts or measurement choices.

4 Results

5 Results

We demonstrate our party-specific overlapping district system through two test cases: Ohio (15 congressional seats) and California (52 seats). These states represent moderate-sized and large delegations, respectively, with different partisan compositions.

5.1 Ohio: Moderate-Sized Delegation

Ohio's 15 congressional seats provide an ideal test case for our system. Using 2020 presidential election results (Biden 45.3%, Trump 53.2%, Other 1.5%), we apply the D'Hondt method with the natural threshold of $1/(N+1) = 6.25\%$.

Seat Allocation The proportional allocation yields 7 Democratic seats and 8 Republican seats, closely matching the statewide vote shares (46.7% vs 53.3% of seats for actual votes of 45.3% vs 53.2%). The "Other" category falls below the threshold and receives no seats.

Party-Weighted Districts We generate two independent sets of overlapping districts—one for Democrats, one for Republicans—each covering the entire state. Critically, districts are balanced by party-vote weighted population rather than total population:

- **Democratic districts**: 7 districts, each containing approximately 763,000 Democratic-weighted residents (Ohio's 11.8M population $\times$ 45.3% $\div$ 7 districts)
- **Republican districts**: 8 districts, each containing approximately 785,000 Republican-weighted residents (11.8M $\times$ 53.2% $\div$ 8 districts)

This weighting ensures equal representation *within* each party's caucus. Every Democratic voter has an equal say in selecting Democratic representatives, regardless of whether they live in Cleveland or rural Appalachia.

Comparison to Traditional Approach Under traditional single-member districts balanced by total population, each district would contain approximately 787,000 residents. However, this creates severe imbalances in party representation. A Democratic district in urban Cleveland might contain 560,000 Democratic voters, while a Democratic district forced into rural areas might contain only 330,000—a 1.7$\times$ imbalance that dilutes rural Democratic voters' representation within the Democratic caucus.

Our party-weighted approach eliminates this imbalance entirely. Table 1 shows the variation in Democratic votes across the 7 Democratic districts ranges only 392,725 to 393,386 (0.17% variation)—effectively perfect balance.

Table 1: Ohio Democratic districts: Party-vote balance

District	Total Pop.	Dem. Votes	Deviation
District 1	1,684,925	392,928	-0.04%
District 2	1,685,847	393,140	+0.01%
District 3	1,686,321	393,259	+0.04%
District 4	1,686,871	393,386	+0.07%
District 5	1,685,594	393,098	0.00%
District 6	1,685,864	393,147	+0.01%
District 7	1,684,026	392,725	-0.10%
Average	1,685,635	393,098	—
Std. Dev.	1,012	226	0.06%

5.2 California: Large Delegation with Third-Party Representation

California's 52-seat delegation demonstrates the system's scalability and its capacity to represent minor parties. With 2020 results (Biden 63.4%, Trump 34.3%, Other 2.3%), the natural threshold of $1/(52+1) = 1.89\%$ allows the "Other" category to qualify for representation.

Seat Allocation The D'Hondt allocation yields:

- 33 Democratic seats (63.5% of seats for 63.4% of votes)
- 18 Republican seats (34.6% of seats for 34.3% of votes)
- 1 Other seat (1.9% of seats for 2.3% of votes)

Three-Party Overlapping Districts California now has *three* independent district maps covering the state:

- **33 Democratic districts**: Average 759,613 Democratic-weighted residents per district
- **18 Republican districts**: Average 753,422 Republican-weighted residents per district
- **1 Other district**: Represents all 39.5M California residents for third-party voters (at-large)

The single "Other" seat functions as an at-large district representing the entire state, giving voice to Libertarians, Greens, and independent voters who would otherwise have no representation.

Proportionality Achievement Table 2 compares our system to California's actual 2022 congressional delegation. Our system achieves near-perfect proportionality (efficiency gap $< 0.1\%$) by construction, while the actual districts exhibit typical partisan bias.

Table 2: Proportionality comparison: Our system vs. actual California districts

System	Vote %	Seats	Seat %	Eff. Gap
Our System (2020)				
Democratic	63.4%	33	63.5%	
Republican	34.3%	18	34.6%	0.08%
Other	2.3%	1	1.9%	
Actual Districts (2022)				
Democratic	—	42	80.8%	15-20%
Republican	—	10	19.2%	

5.3 Key Findings

Our results demonstrate three critical properties of party-specific overlapping districts:

Perfect Internal Balance Within each party's delegation, every representative serves an equal number of their party's voters (standard deviation $< 0.2\%$). This contrasts sharply with traditional districts, where geographic constraints force party representatives to serve vastly different numbers of co-partisans depending on urban/rural geography.

Guaranteed Proportionality By allocating seats *before* drawing districts, our system achieves proportionality by construction. The efficiency gap approaches zero, limited only by rounding constraints in seat allocation.

Third-Party Representation In large delegations, the natural threshold becomes low enough for significant third parties to win representation. California's 1.89% threshold enabled "Other" voters to elect a representative despite comprising only 2.3% of the electorate—impossible under traditional single-member plurality rules.

6 Discussion

7 Discussion

Our party-specific overlapping district system fundamentally reconceptualizes congressional representation by separating proportionality from geography. Rather than forcing geographic districts to achieve proportional outcomes—an often impossible constraint—we achieve proportionality through seat allocation and use districts to ensure equal representation *within* each party.

7.1 Representational Quality

Dual Representation Every citizen gains dual representation: a party-specific representative elected by co-partisans, and implicit representation through their party's proportional seat allocation. A Democrat in rural Kentucky and a Democrat in San Francisco both have equal influence in selecting Democratic representatives, while the overall 7D-8R allocation reflects Kentucky's partisan balance.

This differs fundamentally from traditional systems where geographic accidents determine representation quality. A Democrat in a 70%-Republican district has virtually no influence on representative selection, despite comprising 30% of the district. Our system ensures every voter has equal influence within their chosen party.

Within-Party Fairness Traditional redistricting discourse focuses on fairness *between* parties—eliminating partisan gerrymandering through efficiency gaps or proportional outcomes. We demonstrate equal importance of fairness *within* parties. When urban and rural co-partisans have vastly different numbers of voters per representative, internal party factions gain unequal influence.

Ohio's results illustrate this starkly: traditional balanced districts create a 1.7× imbalance in Democratic representation based on geography. Our party-weighted approach eliminates this, ensuring every Democratic voter has equal say in Democratic representative selection regardless of residence.

7.2 Comparison to Alternative Systems

Proportional Representation Our system achieves PR-like proportionality while maintaining single-member districts and geographic ties. Unlike party-list PR, voters elect specific representatives accountable to geographic constituencies. Unlike mixed-member PR (as in Germany or New Zealand), we eliminate the distinction between constituency and list MPs—all representatives serve geographic districts.

The key innovation is overlapping rather than exclusive geography. Multiple representatives can serve overlapping regions, each accountable to different partisan constituencies.

Ranked-Choice Voting and Multi-Member Districts RCV in single-winner elections cannot guarantee proportionality without multi-member districts. Multi-member RCV (STV) achieves proportionality but requires large constituencies that sacrifice geographic representation. Our system maintains small, geographically meaningful districts while achieving proportionality through overlapping coverage.

Independent Redistricting Commissions Commissions reduce partisan gerrymandering but cannot guarantee proportionality without sacrificing traditional districting criteria (compactness, contiguity, community preservation). Our system achieves proportionality by design while maintaining algorithmic objectivity—no commission discretion required.

7.3 Practical Implementation

Ballot Design Voters would receive a party-specific ballot listing only candidates from their declared party preference (or all candidates for independents choosing the nonpartisan ballot). This simplifies rather than complicates voting—most voters already identify with a party and prefer choosing among co-partisans.

Primary elections become less critical since general elections now serve the function of intra-party choice. States could eliminate or consolidate primaries, reducing election costs and voter fatigue.

Representative Duties Representatives serve geographically-defined districts but are elected by party-specific constituencies. A Democratic representative covers specific census tracts, responding to all constituent service requests in that geography while being accountable to Democratic voters for re-election.

This clarifies rather than complicates representation. Current representatives claim to serve "all constituents" while being elected by and accountable to only their party's voters. Our system makes this reality explicit while ensuring proportional voice for all parties.

Third-Party Viability The natural threshold formula creates different representation possibilities across delegation sizes. Small states (2-4 seats) require 20-33% vote share for third-party representation—difficult but not impossible. Large states (California's 1.89% threshold) enable meaningful third-party representation.

This gradualism is arguably optimal: minor parties must demonstrate substantial support to gain representation, but the barrier is achievable. California's "Other" seat, representing 900,000 voters (2.3% of 39.5M), provides more representation than those voters receive under current plurality rules.

7.4 Limitations and Extensions

Data Requirements Our implementation uses statewide vote disaggregation to estimate tract-level party support. Real deployment would benefit from precinct-level data aggregated to tracts, capturing geographic variation in party support. However, our results demonstrate the system's viability even with simplified data.

Nonpartisan and Independent Voters Our current model allocates seats only to parties receiving votes. Future work should consider whether nonvoters or registered independents deserve guaranteed representation through at-large districts or reserved seats. If 30% of eligible voters abstain, does proportional representation require explicitly representing their interests?

Compactness Trade-offs Party-weighted districts prioritize party-vote balance over total-population balance, potentially creating less compact districts than traditional approaches. However, our recursive bisection algorithm still optimizes compactness within the constraint of party-vote balance. Empirical analysis of resulting district shapes remains future work.

Dynamic Effects How would this system affect political behavior? Would party switching become more common? Would parties splinter knowing representation is proportional? Would gerrymandering shift to manipulating party registration? These questions require modeling beyond this initial implementation.

7.5 Normative Implications

Our system embodies a specific theory of representation: political parties, not geography, are the primary axis of political identity and interest aggregation in modern democracies. Geographic representation matters for constituent services and local accountability, but proportional representation matters more for legislative outcomes.

This challenges American exceptionalism in clinging to single-member plurality districts. Most advanced democracies use proportional or mixed systems. Our approach offers a path to proportionality while preserving the geographic representation Americans value.

The overlapping districts concept may seem radical, but it formalizes existing reality. Congressional districts already overlap with state legislative districts, county boundaries, and municipal governments. Citizens already navigate multiple overlapping governmental jurisdictions. Adding party-specific congressional districts simply makes explicit the partisan nature of representation while achieving proportionality impossible under traditional constraints.

8 Conclusion

9 Conclusion

This paper demonstrates that perfect proportional representation is achievable in single-member district systems through party-specific overlapping districts. By separating the problem of proportional seat allocation from the problem of geographic districting, we escape the fundamental impossibility of achieving both proportionality and traditional districting criteria simultaneously.

Our key insight is that fairness operates on two dimensions: fairness *between* parties (proportionality) and fairness *within* parties (equal representation per voter). Traditional redistricting sacrifices one or both. Single-member plurality districts sacrifice proportionality. Proportional representation systems sacrifice within-party geographic balance or abandon single-member districts entirely. Our system achieves both by allowing each party to independently draw districts balanced by party-specific populations.

Ohio and California demonstrate practical viability. Moderate delegations achieve two-party proportionality with perfect internal balance. Large delegations enable third-party representation through low natural thresholds. The algorithmic approach requires no discretionary commission decisions—proportionality and balance emerge from transparent mathematical procedures.

The most radical element—overlapping districts—is less radical than it appears. Citizens already navigate overlapping jurisdictions (federal, state, local). Representatives already serve partisan rather than geographic constituencies in practice. Our system makes explicit what traditional districts obscure: in polarized democracies, party matters more than place for representation quality.

9.1 Future Research

Several extensions merit investigation:

Empirical Analysis Full 50-state implementation would reveal how delegation size affects representation quality, whether

geographic constraints bind differentially across regions, and how compactness trades off against party-vote balance in practice.

Dynamic Effects Behavioral modeling could illuminate how parties, voters, and candidates adapt to proportional rules. Would parties fragment knowing representation is guaranteed? Would gerrymandering shift to manipulating party registration? Would primary elections retain importance?

Comparative Systems How does our approach compare empirically to mixed-member PR, ranked-choice voting in multi-member districts, or independent commission redistricting? Systematic comparison across metrics (proportionality, geographic representation, voter satisfaction, policy outcomes) would clarify relative advantages.

Constitutional Viability What legal barriers exist to implementing overlapping districts? Does the Constitution's requirement that representatives be "apportioned among the several States" preclude this approach? Could states adopt party-specific districts for state legislatures without federal constitutional amendment?

Nonpartisan Representation Should nonvoters or independents receive guaranteed representation through at-large seats or reserved districts? How should we balance parties' proportional claims against nonpartisans' representation interests?

9.2 Closing Thoughts

American democracy faces a crisis of representation. Gerrymandering undermines fairness. Winner-take-all districts waste votes. Geographic sorting makes competitive districts scarce. Our system offers a path forward: achieve proportionality through seat allocation, achieve intra-party fairness through party-weighted districts, and maintain geographic representation through overlapping coverage.

The approach is technically feasible—our implementation proves this. The question is whether political will exists to reconceptualize representation around parties rather than exclusive geography. For a nation that accepts overlapping school districts, water districts, and municipal boundaries, overlapping congressional districts may be less radical than clinging to 18th-century assumptions about representation in a 21st-century partisan environment.